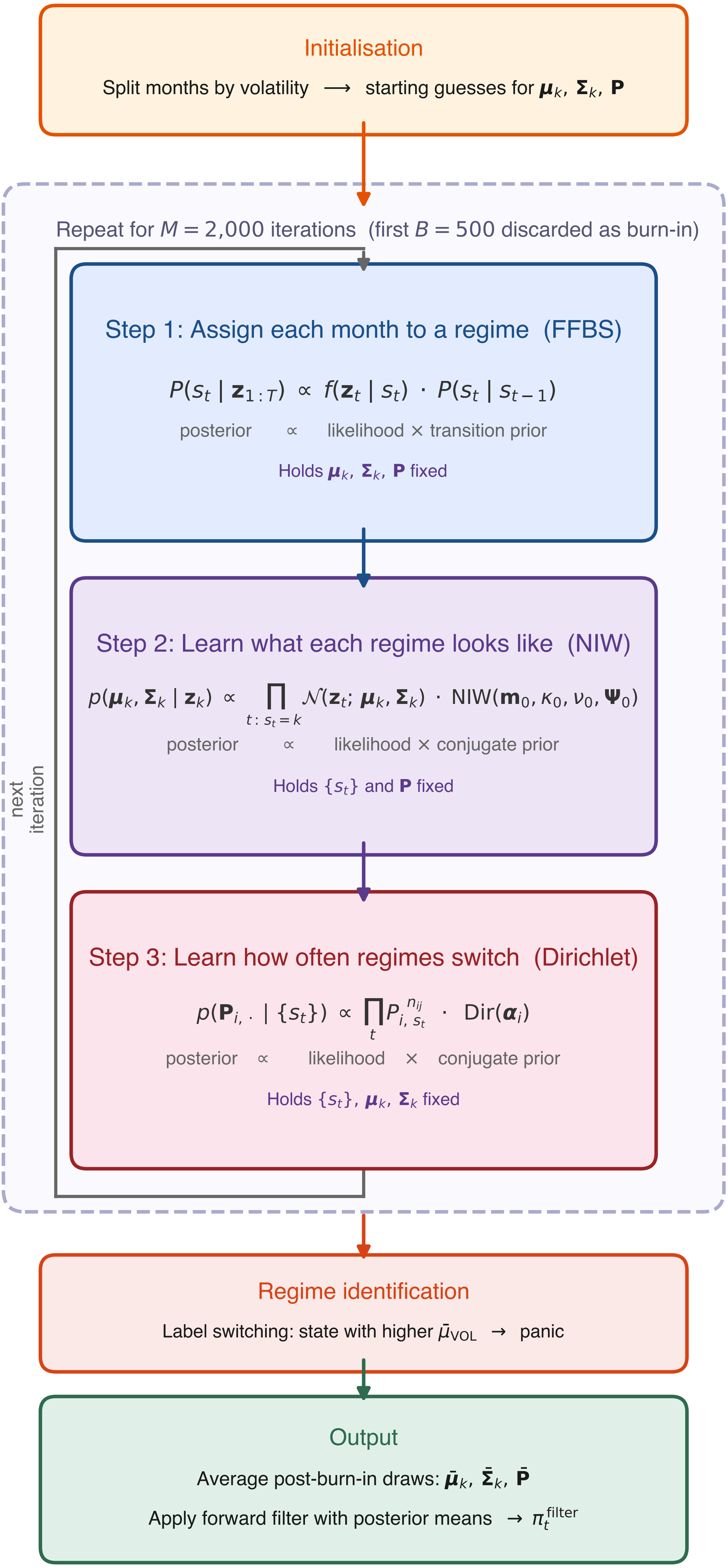
Initialisation
Split months by volatility $\longrightarrow$ starting guesses for $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$
Repeat for $M = 2{,}000$ iterations (first $B = 500$ discarded as burn-in)
Step 1: Assign each month to a regime (FFBS)
$P(s_t \mid \mathbf{z}_{1:T}) \propto f(\mathbf{z}_t \mid s_t) \cdot P(s_t \mid s_{t-1})$
posterior $\propto$ likelihood × transition prior
Holds $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$, $\mathbf{P}$ fixed
Step 2: Learn what each regime looks like (NIW)
$p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k \mid \mathbf{z}_k) \propto \prod_{t:\, s_t = k} \mathcal{N}(\mathbf{z}_t; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \cdot \mathrm{NIW}(\mathbf{m}_0, \kappa_0, \nu_0, \boldsymbol{\Psi}_0)$
posterior $\propto$ likelihood × conjugate prior
Holds $\{s_t\}$ and $\mathbf{P}$ fixed
Step 3: Learn how often regimes switch (Dirichlet)
$p(\mathbf{P}_{i,\cdot} \mid \{s_t\}) \propto \prod_t P_{i,\, s_t}^{n_{ij}} \cdot \mathrm{Dir}(\boldsymbol{\alpha}_i)$
posterior $\propto$ likelihood × conjugate prior
Holds $\{s_t\}$, $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ fixed
next iteration
Regime identification
Label switching: state with higher $\bar{\mu}_{\mathrm{VOL}}$ $\rightarrow$ panic
Output
Average post-burn-in draws: $\bar{\boldsymbol{\mu}}_k$, $\bar{\boldsymbol{\Sigma}}_k$, $\bar{\mathbf{P}}$
Apply forward filter with posterior means $\rightarrow \pi_t^{\mathrm{filter}}$